

Tracking the popularity of baby names in the United States

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This file is our first look at R. You will follow the instructions in this file, and at the end, you will upload the end-product (a PDF file) to Blackboard for grading.

To see what this file does, click the “Knit” button just above this display. When you look through the file, you will see that I have given you three jobs to customize the output for yourself.

Task 1

Change the name of the author at the top of this file to your name, and the date to today.

Introduction

How has the popularity of various baby names in the US changed over time? We will explore this question using records of first names supplied by the US Social Security Administration. This government organization has a complete, publically available database of the first names for everyone with birth year 1880 or later who has applied for a Social Security number.

Packages

We will use the **tidyverse** and **scales** packages for data wrangling and visualization.

```
library(tidyverse)
library(scales)
```

Data

That data that we’re using originally come from the Social Security Administration. This particular data file was prepared for the textbook *Regression and Other Stories* (Gelman *et al*) and comes from the data repository for the book. (<https://github.com/avehtari/ROS-Examples/tree/master/Names/>)

The following complicated chunk loads the data file and reorganizes it. The code also calculates how common each name is within the records for each year. You will not have to do anything with this!

```
names <- read_csv(file = "data/SSA-longtail-names.csv",
                  col_types = cols(sex = col_character())) %>%
  select(-...1) %>%
  pivot_longer(cols = "1880":"2010",
               names_to = "year",
               values_to = "count") %>%
  mutate(year = as.numeric(year)) %>%
  group_by(sex, year) %>%
  mutate(prop_by_year = count / sum(count)) %>%
  ungroup()
```

```
## New names:
## * `` -> `...1`
```

Ranks of names for a particular year

Here is a table of the 100 most popular girls' names for the birth year 1981. (changed to most popular boys names for 1995)

```
top_names <- names %>% filter(year == 1995, sex == "M") %>%
  select(name, count) %>%
  slice_max(order_by = count, n = 100)
print(top_names, n = 100)
```

```
## # A tibble: 100 x 2
##   name      count
##   <chr>    <dbl>
## 1 Michael  41396
## 2 Matthew  32870
## 3 Christopher 32662
## 4 Jacob    31106
## 5 Joshua   30711
## 6 Nicholas 29156
## 7 Tyler    29148
## 8 Brandon  26891
## 9 Daniel   26679
## 10 Austin  25903
## 11 Andrew  25826
## 12 Joseph  25323
## 13 John    23209
## 14 Zachary 23131
## 15 Ryan    23088
## 16 David   23014
## 17 James   22717
## 18 Justin  21575
## 19 Anthony 20478
## 20 William 20142
## 21 Alexander 19445
## 22 Robert  18576
## 23 Jonathan 18141
## 24 Kyle    16760
## 25 Kevin   15847
## 26 Cody    15564
## 27 Thomas  14160
## 28 Christian 13801
## 29 Jordan  13564
## 30 Aaron   13277
## 31 Benjamin 12698
## 32 Eric    12474
## 33 Samuel  12404
## 34 Dylan   12089
## 35 Jose    11774
## 36 Brian   10985
## 37 Steven  10658
## 38 Adam    10479
```

##	39 Nathan	10303
##	40 Timothy	9780
##	41 Jason	9001
##	42 Richard	8778
##	43 Patrick	8672
##	44 Charles	8473
##	45 Sean	8164
##	46 Jesse	7483
##	47 Alex	7422
##	48 Jeremy	7084
##	49 Cameron	7029
##	50 Juan	6980
##	51 Logan	6951
##	52 Mark	6882
##	53 Hunter	6713
##	54 Caleb	6648
##	55 Connor	6616
##	56 Dakota	6576
##	57 Stephen	6550
##	58 Devin	6184
##	59 Luis	6164
##	60 Evan	6088
##	61 Trevor	6008
##	62 Dustin	5920
##	63 Jared	5881
##	64 Jeffrey	5729
##	65 Gabriel	5710
##	66 Travis	5603
##	67 Ian	5594
##	68 Carlos	5540
##	69 Taylor	5519
##	70 Bryan	5441
##	71 Paul	5410
##	72 Kenneth	5227
##	73 Nathaniel	5201
##	74 Corey	4975
##	75 Ethan	4934
##	76 Jesus	4904
##	77 Blake	4819
##	78 Bradley	4765
##	79 Mitchell	4734
##	80 Garrett	4710
##	81 Marcus	4641
##	82 Luke	4625
##	83 Derek	4452
##	84 Gregory	4442
##	85 Antonio	4337
##	86 Tanner	4319
##	87 Dalton	4190
##	88 Edward	4136
##	89 Peter	4109
##	90 Victor	4093
##	91 Scott	4084
##	92 Lucas	4082

```
## 93 Elijah      3976
## 94 Miguel      3948
## 95 Chase       3916
## 96 Adrian     3913
## 97 Seth        3912
## 98 Isaac       3876
## 99 Shawn       3820
## 100 Noah       3817
```

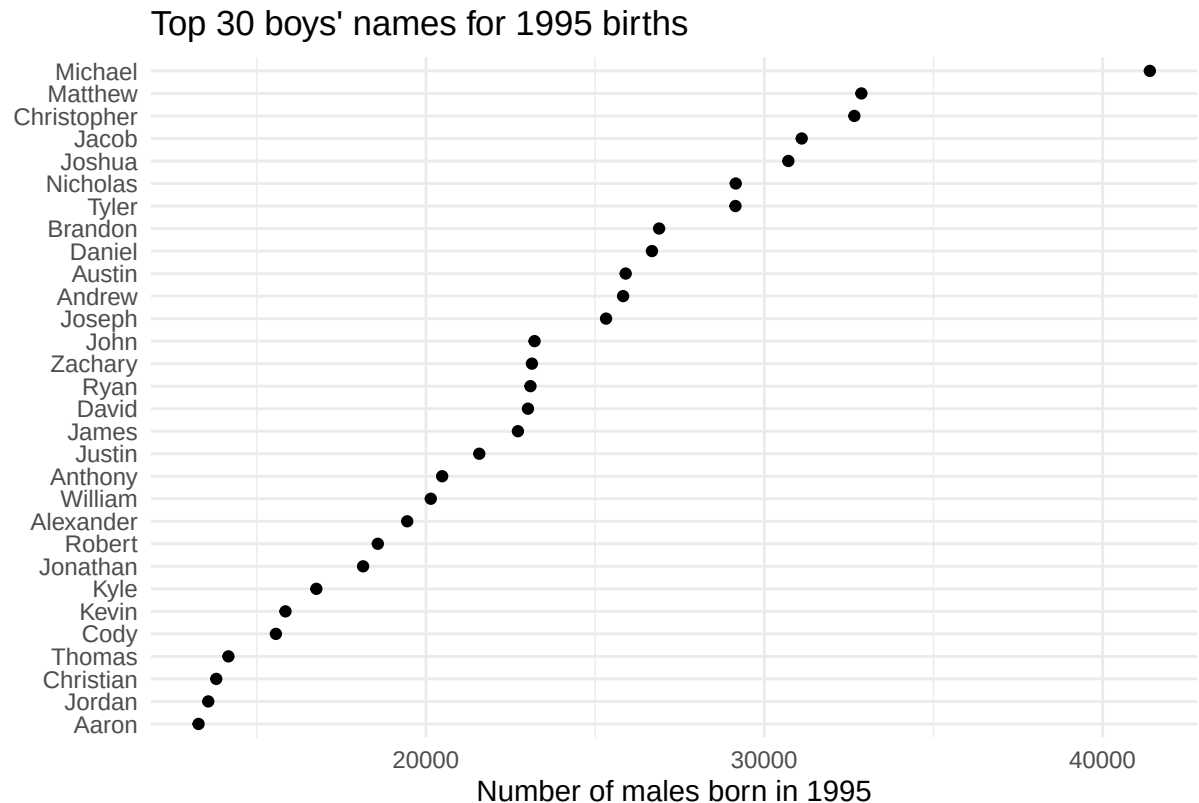
This chunk pulls out the row for the name **Sara** and tells us the rank of the name in order of popularity for that year. It would work even if **Sara** were not in the top 100 names for that year.

```
names %>% filter(year == 1995, sex == "M") %>%
  select(name, count) %>%
  mutate(rank = dense_rank(desc(count))) %>%
  filter(name == "Daniel")
```

```
## # A tibble: 1 x 3
##   name    count rank
##   <chr>   <dbl> <int>
## 1 Daniel 26679     9
```

The following chunk makes a graph showing the popularity of the top 30 girls' names for 1981 births.

```
ggplot(data = top_names %>% slice_max(order_by = count, n = 30),
  mapping = aes(x = count, y = fct_reorder(name, count))) +
  geom_point() +
  labs(x = "Number of males born in 1995",
    y = "",
    title = "Top 30 boys' names for 1995 births",
    caption = "Source: US Social Security Administration") +
  theme_minimal()
```



Source: US Social Security Administration

Task 2

Go back and change the code in the previous three code chunks to make them mean something to you. For example, you could: * Change 1981 and F in the first chunk to get a table that is more interesting to you. * You could change the second chunk similarly to match yourself or someone you know. * In the third chunk, you should change the title and x-axis label appropriately to reflect what you did in the first chunk.

Visualization of trends over time.

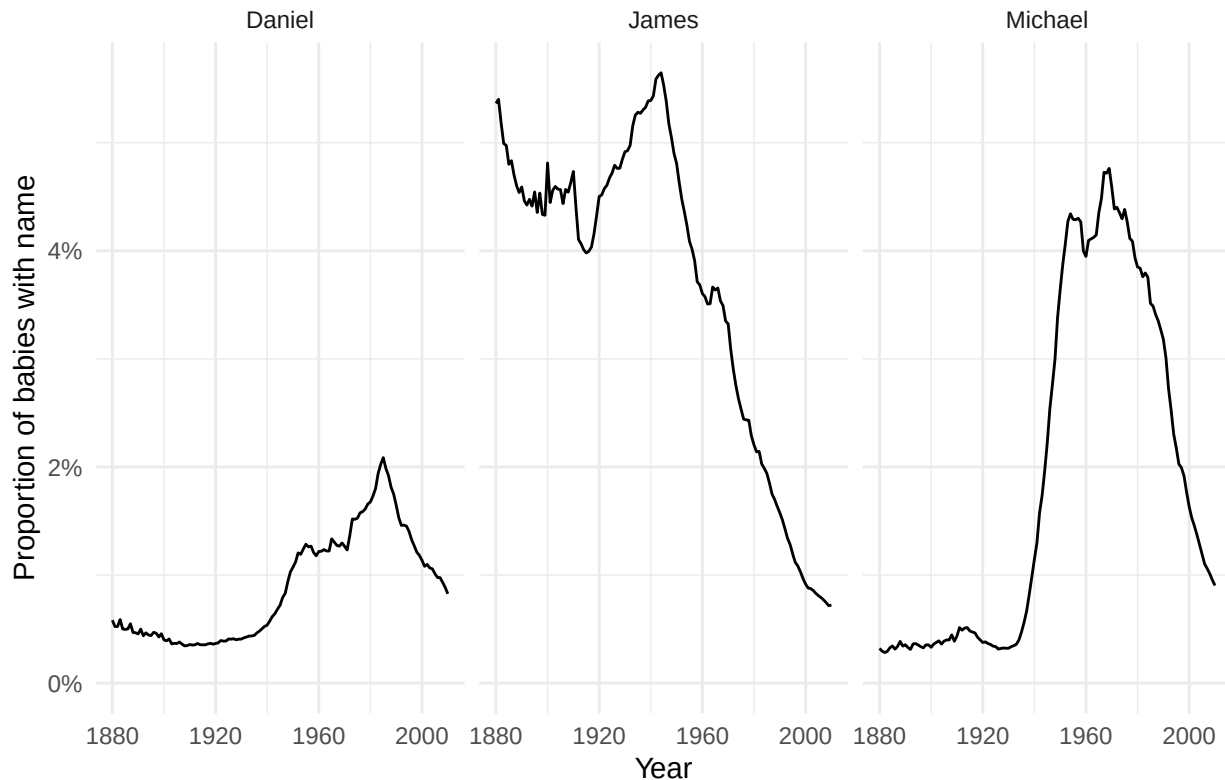
Next, instead of focusing on the particular year 1981, we will look at trends in the popularity of names over time by making some graphs. I have picked out three names that are meaningful to me. To keep things simple, I am looking at name records identified as female.

```
my_names <- names %>% filter(sex == "M",
                             name %in% c("Daniel", "Michael", "James"))
```

The next chunk makes graphs showing the relative popularity of these names over time. The story is quite different for each of my three names!

```
ggplot(data = my_names, mapping = aes(x = year, y = prop_by_year)) +
  geom_line() +
  facet_wrap(~name) +
  scale_y_continuous(labels = percent, limits = c(0, NA)) +
  labs(x = "Year", y = "Proportion of babies with name",
       title = "Trends in boys name popularity over time") +
  theme_minimal()
```

Trends in boys name popularity over time



Task 3

Change the code in the first code chunk of this section to make it mean something to you. You should follow my lead and restrict your attention either only to M records or only to F records. You can choose as many names to compare as you want. Once you see the graphs, you may want to adjust your choices. *After you have a nice set of graphs, write a few sentences in the space below describing what you see.*

Write your sentences here:

In the graphs above, I observe that the name “Daniel” shows a peak in popularity during the late 1980s to early 1990s, followed by a gradual decline just before the 2000s. Despite this decline, “Daniel” has remained a relatively consistent name over time.

Additionally, the name “James” was one of the most popular names for boys from 1880 to the late 1950s. After its peak during this time frame, its popularity sharply declined until the 2000s, where it has become less common, now falling below the 2% mark.

Furthermore, the name “Michael” started with a low popularity rate (just below 1%) until the 1940s. Between the 1940s and 1960s, it experienced a sharp rise in popularity, continuing to dominate through the late 20th century. However, after the early 2000s, “Michael” has seen a significant decline in popularity in recent years.

In closing, all three names show varying patterns of popularity, reflecting different trends in naming conventions across generations. The decline in popularity of these names since the 2000s may indicate a shift toward more diverse or modern names.

When you are done, make sure to knit your file one more time. Save your file as a PDF to your OneDrive and upload it to Blackboard for grading. This will count towards the Homework and Graded Classwork part of your grade.